



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KPS Fuel Service

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

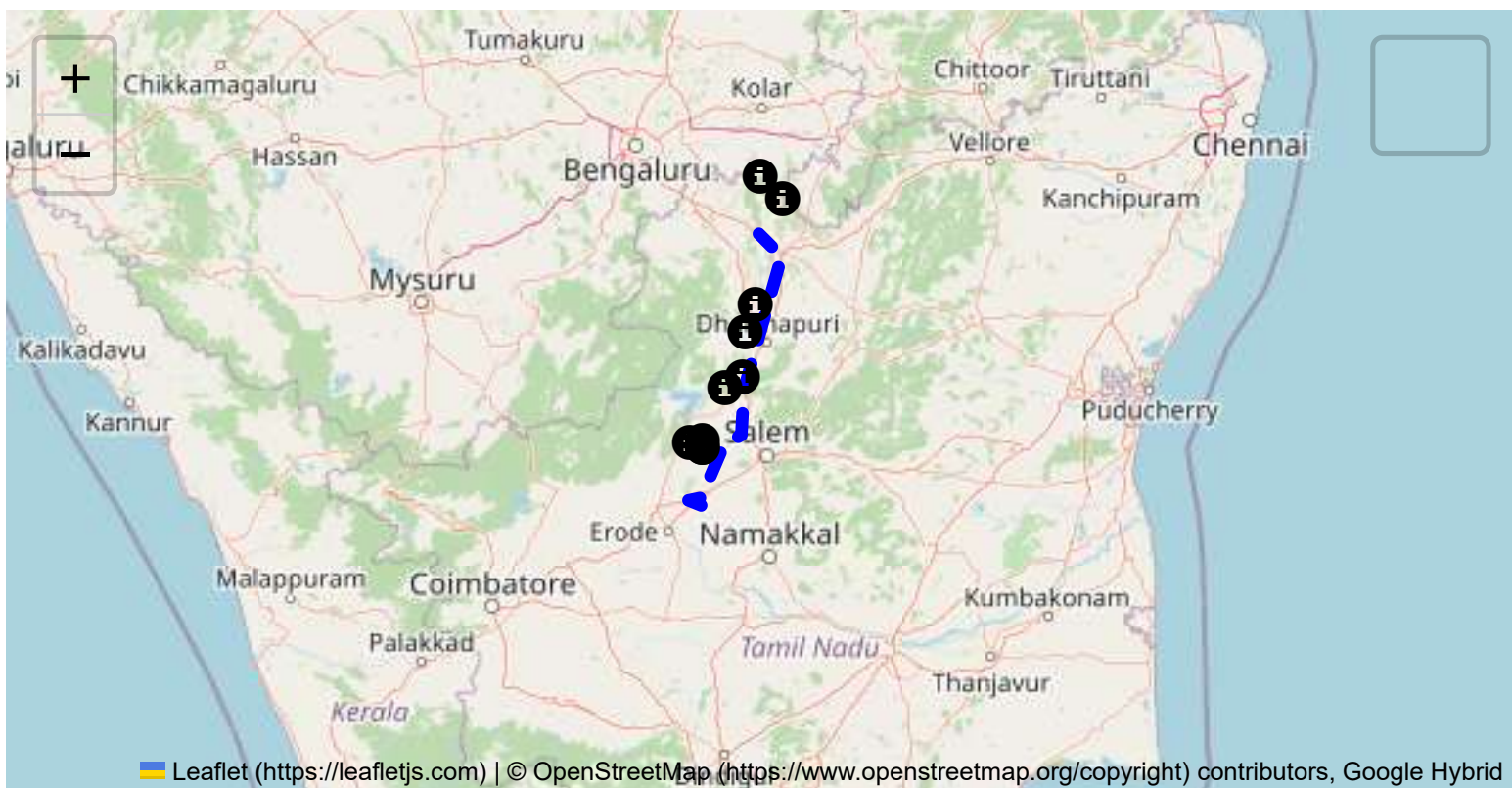
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

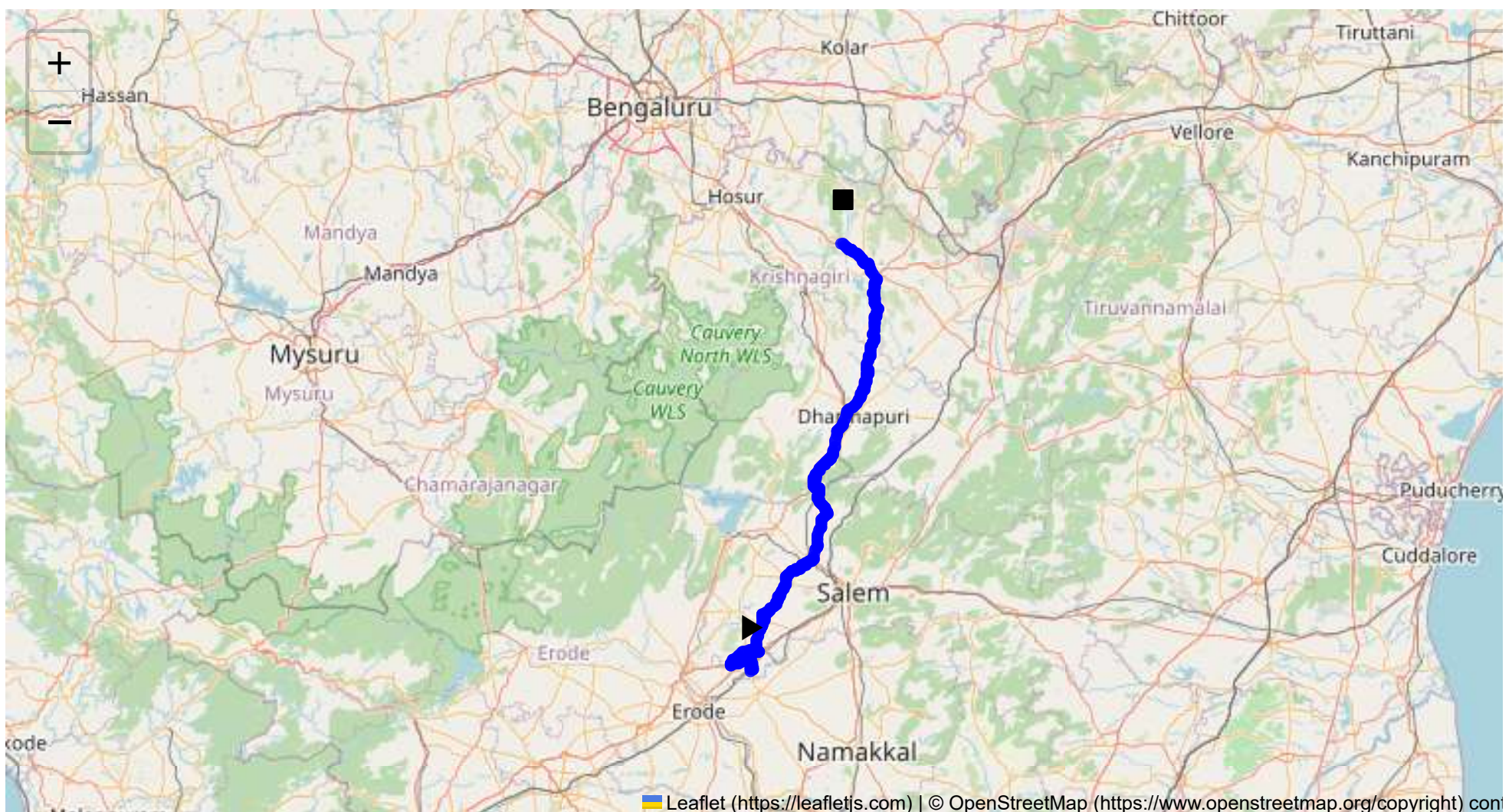
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 170.06 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.601585, 78.124455)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route spans approximately 170 kilometers from Sangagiri to Kurubarapalli, passing through key junctions such as Kangeyampalayam, Padaiveedu, Tharamangalam, Omalur, Thoppur, Seshampatti, and Krishnagiri. It navigates a mix of state highways (SH) and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Weather:** The region experiences a tropical climate. Summers (March to June) can be extremely hot, while the monsoon season (July to September) brings heavy rains that can lead to flooding and reduced visibility.
- **Hazards:** During the monsoon, localized flooding might obstruct some segments of the route. In summer, high temperatures can cause road damage and affect vehicle performance.

3. Analysis of Traffic Patterns:

- **Peak Hours:** Typically, peak traffic occurs during morning hours (8-10 AM) and early evenings (5-7 PM), especially near towns and major intersections like Salem and Krishnagiri.
- **Congestion Areas:** Traffic congestion is common in urban areas and near major junctions, such as in Omalur and Krishnagiri.

4. Assessment of Road Quality and Infrastructure:

- The road quality varies, with well-maintained highways and some sections with potholes or narrower lanes, especially local roads. Infrastructure is generally adequate, but be cautious of potential construction zones.

5. Suggestions for Alternative Routes for Emergencies:

- An alternative pathway could involve taking NH44 through Salem and Dharmapuri, then rerouting to the intended destination. This national highway often provides better maintained and clearer pathways during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Tamil Nadu requires permits for hazardous material transport, and specific routes are designated to avoid populated areas, especially during the night. Compliance with the Motor Vehicles Act concerning signage and vehicle conditions is essential.

7. Overview of Historical Incidents:

- Heavy vehicle incidents are often reported on this corridor due to driver fatigue and speeding. Known accident-prone zones include Salem-Krishnagiri stretch.

8. Environmental Considerations and Sensitive Areas:

- The route passes near environmentally sensitive areas, including agricultural land and a few densely wooded sections. Caution is advised to prevent contamination from spills or littering.

9. Analysis of Communication Coverage:

- The route mostly has good mobile network coverage, with potential dead zones in remote sections between Omalur and Thoppur. Ensure to have a backup communication plan.

10. Estimated Emergency Response Times:

- In cities and towns like Salem and Krishnagiri, response times can be quicker (about 30 minutes), whereas rural segments may experience longer delays (up to an hour or more).

11. Overall Risk Assessment Summary:

- The route is moderately risky with concerns mainly during adverse weather and congested urban areas. Adequate preparation, including knowledge of local laws, familiarization with emergency routes, and communication readiness, can mitigate most risks. It is advised for drivers to

maintain strict adherence to speed limits, particularly in areas known for vehicle incidents. Regular vehicle maintenance checks are crucial to ensure safety, especially in challenging conditions like high temperatures or heavy rains.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4749755, 77.869149	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44077, 77.87533	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	Medium	11.46392, 77.87328	30 KM/Hr
6	Turn	Medium	11.46755, 77.87044	30 KM/Hr
7	Turn	High	11.47391, 77.86844	15 KM/Hr
8	Blind Spot	Blind Spot	11.47481, 77.86920	10 KM/Hr
9	Turn	Medium	11.47354, 77.86775	30 KM/Hr
10	Turn	Medium	11.45543, 77.81780	30 KM/Hr
11	Blind Spot	Blind Spot	11.45615, 77.81677	10 KM/Hr
12	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
13	Turn	Medium	11.49377, 77.88573	30 KM/Hr
14	Turn	High	11.49469, 77.88488	15 KM/Hr
15	Turn	Medium	11.55618, 77.89789	30 KM/Hr
16	Turn	Medium	11.55699, 77.89986	30 KM/Hr
17	Turn	High	11.58228, 77.90841	15 KM/Hr
18	Turn	High	11.58340, 77.90637	15 KM/Hr
19	Turn	Medium	11.59201, 77.90649	30 KM/Hr
20	Turn	Medium	11.63882, 77.94859	30 KM/Hr
21	Turn	High	11.64226, 77.95200	15 KM/Hr
22	Turn	Medium	11.64716, 77.95118	30 KM/Hr
23	Turn	Medium	11.69553, 77.97110	30 KM/Hr
24	Turn	Medium	11.69806, 77.97537	30 KM/Hr
25	Turn	Medium	11.71386, 77.99552	30 KM/Hr
26	Turn	Medium	11.71735, 78.00019	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
27	Turn	Medium	11.71915, 78.01106	30 KM/Hr
28	Turn	Medium	11.72492, 78.01660	30 KM/Hr
29	Turn	Medium	11.72765, 78.02291	30 KM/Hr
30	Turn	Medium	11.73764, 78.03441	30 KM/Hr
31	Turn	Medium	11.73752, 78.03679	30 KM/Hr
32	Turn	High	11.73652, 78.03866	15 KM/Hr
33	Turn	Medium	11.74110, 78.04575	30 KM/Hr
34	Turn	Medium	11.74340, 78.04719	30 KM/Hr
35	Turn	Medium	11.77984, 78.05782	30 KM/Hr
36	Turn	Medium	11.86902, 78.08380	30 KM/Hr
37	Turn	Medium	11.87988, 78.08176	30 KM/Hr
38	Turn	Medium	11.92637, 78.06515	30 KM/Hr
39	Turn	Medium	11.93180, 78.06478	30 KM/Hr
40	Turn	Medium	11.93467, 78.06277	30 KM/Hr
41	Turn	Medium	11.93726, 78.05359	30 KM/Hr
42	Turn	Medium	11.95482, 78.04687	30 KM/Hr
43	Turn	High	11.95837, 78.04986	15 KM/Hr
44	Turn	High	11.96007, 78.04945	15 KM/Hr
45	Turn	High	11.96415, 78.05126	15 KM/Hr
46	Turn	Medium	11.96519, 78.04958	30 KM/Hr
47	Turn	High	11.96620, 78.04920	15 KM/Hr
48	Turn	Medium	11.96842, 78.05097	30 KM/Hr
49	Turn	Medium	12.17005, 78.16792	30 KM/Hr
50	Turn	Medium	12.19919, 78.18172	30 KM/Hr
51	Turn	Medium	12.22472, 78.19140	30 KM/Hr
52	Turn	Medium	12.24623, 78.19524	30 KM/Hr
53	Turn	Medium	12.25078, 78.19988	30 KM/Hr
54	Turn	Medium	12.25370, 78.20034	30 KM/Hr
55	Turn	Medium	12.29150, 78.19930	30 KM/Hr
56	Turn	Medium	12.39930, 78.21762	30 KM/Hr
57	Turn	Medium	12.40604, 78.22285	30 KM/Hr
58	Turn	Medium	12.42500, 78.22672	30 KM/Hr
59	Turn	Medium	12.50188, 78.22338	30 KM/Hr
60	Turn	Medium	12.50577, 78.22309	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
61	Turn	Medium	12.59755, 78.13586	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
7	hospital	SP Hospital	11.4708567, 77.8654872	30 km/h	Medium
8	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
9	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
10	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
12	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
13	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
14	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
15	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
16	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
17	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
18	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
19	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
20	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
21	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
22	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
24	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
25	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
26	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
27	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
28	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
29	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
31	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
32	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
34	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
36	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
37	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
38	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
39	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
44	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
45	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
46	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
48	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
49	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
50	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
11	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
33	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
35	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
40	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
41	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
42	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
43	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
47	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
51	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4749755, 77.869149



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46392, 77.87328



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46755, 77.87044



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47391, 77.86844



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.47481, 77.86920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47354, 77.86775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81780



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63882, 77.94859



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64226, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64716, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69553, 77.97110



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69806, 77.97537



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71915, 78.01106



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72492, 78.01660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73764, 78.03441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73752, 78.03679



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74110, 78.04575



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74340, 78.04719



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77984, 78.05782



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86902, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87988, 78.08176



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92637, 78.06515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93180, 78.06478



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93467, 78.06277



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93726, 78.05359



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95482, 78.04687



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95837, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96007, 78.04945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96415, 78.05126



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96519, 78.04958



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96620, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96842, 78.05097



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17005, 78.16792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.19919, 78.18172



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22472, 78.19140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24623, 78.19524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25078, 78.19988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25370, 78.20034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29150, 78.19930



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.39930, 78.21762



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40604, 78.22285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.42500, 78.22672



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50188, 78.22338



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50577, 78.22309



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.59755, 78.13586
